Example 6.8 Find the equation of a straight line cutting an intercept of 5 from the negative direction of the y-axis and is inclined at an angle 150° to the x-axis.

Solution:

Given that the negative y intercept is 5 i.e., $b = -5$ and $\theta = 150°$

slope $m = \tan 150° = \tan(180° - 30°) = -\tan 30° = -\dfrac{1}{\sqrt{3}}$

Slope and intercept form of the equation is $y = mx + b$.

That is $y = -\dfrac{1}{\sqrt{3}}x - 5$.

$$x + \sqrt{3}y + 5\sqrt{3} = 0$$

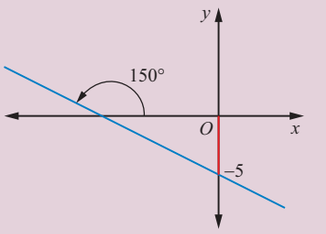


Figure 6.26

Example 6.9 Show the points $\left(0, -\dfrac{3}{2}\right)$, $(1, -1)$ and $\left(2, -\dfrac{1}{2}\right)$ are collinear.

Solution:

Let A, B and C be $\left(0, -\dfrac{3}{2}\right)$, $(1, -1)$ and $\left(2, -\dfrac{1}{2}\right)$ respectively.

The slope of AB is $\dfrac{-1 + \frac{3}{2}}{1 - 0} = \dfrac{1}{2}$

The slope of BC is $\dfrac{-\frac{1}{2} + 1}{2 - 1} = \dfrac{1}{2}$

Thus, the slope of AB is equal to slope of BC.
Hence, A, B and C are lying on the same line.

If the rate of change of one variable with respect to the other variable is constant, then this constant rate of change can be taken as slope.(such as speed, constant increase or constant decrease…). Also equations of straight lines depend on the coordinate axes how we define it. Thus in real world problems the equations are not necessarily identical but the path and distance will always be the same.

Example 6.10 The Pamban Sea Bridge is a railway bridge of length about 2065 m constructed on the PalkStrait, which connects the Island town of Rameswaram to Mandapam, the main land of India. The Bridge is restricted to a uniform speed of only 12.5 m/s. If a train of length 560 m starts at the entry point of the bridge from Mandapam, then

(i) find an equation of the motion of the train.
(ii) when does the engine touch island
(iii) when does the last coach cross the entry point of the bridge
(iv) what is the time taken by a train to cross the bridge.